Numerical Analysis Assignments

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Yiping Cheng
ypcheng@bjtu.edu.cn
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School of Electronic and Information Engineering
Beijing Jiaotong University

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Chapter 1 assignment

1. Let $v = -3.5$. Give the binary representation of the 32-bit floating point representation of v.
2. Find the smallest real number that will be represented as $+\infty$ in IEEE 754 double-precision floating point system.
3. What is the machine epsilon for the IEEE 754 double-precision floating point system?
4. Find how the following can be evaluated as accurately as possible in a computer (you can use sinc, and assume it has an accurate implementation):
   1. $f_1(x) = \frac{1}{1+2x} - \frac{1-x}{1+x}$, for $|x| << 1$
   2. $f_2(x) = \sqrt{x + \frac{1}{x}} - \sqrt{x - \frac{1}{x}}$, for $|x| >> 1$
   3. $f_3(x) = \frac{1-\cos x}{x^2}$, for $|x| << 1$
   4. $f_4(x) = \sin(1+x) - \sin 1$, for $|x| << 1$.

5. The number β is said to approximate α to t significant digits if t is the largest nonnegative integer for which

$$\left|\frac{\beta-\alpha}{\alpha}\right| \leq 5 \times 10^{-t}.$$

For each of the following numbers, find its 3-significant-digit representation, and provide your justification:

1. $\frac{7}{13}$
2. $-\frac{500}{9}$

Chapter 2 assignment

1. Suppose bisection is used to solve the equation $x^3 + x - 1 = 0$, with initial bracketing interval $[0, 1]$. Then what are the bracketing intervals when the 1st and 2nd iterations are finished, respectively?
2. If Newton's method is used on $f(x) = x^3 - x + 1$ starting with $x_0 = 1$, what will x_1 be?
3. Write a program in a language such as C, C++, Matlab, Python, Julia, Java, C# or any other well known language, to solve the above equation using Newton's method. The termination condition is $|f(x_k)|$ ceases to decrease.